■ AI PRECISION AGRICULTURE & CROP HEALTH REPORT

Sentinel-2 Satellite Analysis & Machine Learning Agronomic Assessment

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■ Multispectral Vegetation Index Statistics

Vegetation Index	Mean	Min	Max	Median	Std Dev
NDVI (Normalized Veg)	0.1837	0.0662	0.3938	0.1473	0.0803
NDRE (RedEdge Chlorophyll)	0.0970	-0.0113	0.2460	0.0813	0.0607
EVI (Enhanced Veg)	0.1545	0.0709	0.3383	0.1260	0.0665
SAVI (Soil Adjusted)	0.1226	0.0520	0.2716	0.0953	0.0554

■ Vegetation Condition Zone Distribution

Condition Zone	NDVI Threshold	Farm Coverage (%)
Dense / Vigorous Crop Growth (NDVI ≥ 0.50)	NDVI Range	0.0%
Moderate Vegetation Coverage (0.30 ≤ NDVI < 0.50)	NDVI Range	14.23%
Sparse / Emergent Crop Growth (0.15 ≤ NDVI < 0.30)	NDVI Range	33.4%
Bare Soil or Stressed Zone (NDVI < 0.15)	NDVI Range	52.37%

■ AI Machine Learning Crop Health Prediction

ML Health Classification:	**SEVERE_STRESS** (ML model predicts low biomass density and potential vegetation stress risk.)
Health Score:	25.0 / 100
Stress Risk Level:	HIGH_RISK
Model Confidence:	100.0%

■ Actionable Agronomic Recommendations

- **[HIGH] Field Scouting — Inspect Low-Vegetation Zones**

Prioritize ground scouting in the 85.77% low-vegetation sectors to check for moisture deficit, seedling emergence issues, or soil compaction.

- **[MEDIUM] Nutrient & Nitrogen Management — Chlorophyll / Nitrogen Assessment**

  RedEdge NDRE index is below optimal levels. Consider variable-rate nitrogen top-dressing in low NDRE field sectors.

- **[HIGH] Irrigation Management — Soil Moisture Check**

  SAVI response indicates potential soil moisture stress or sparse canopy. Verify irrigation scheduling and soil moisture levels.

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